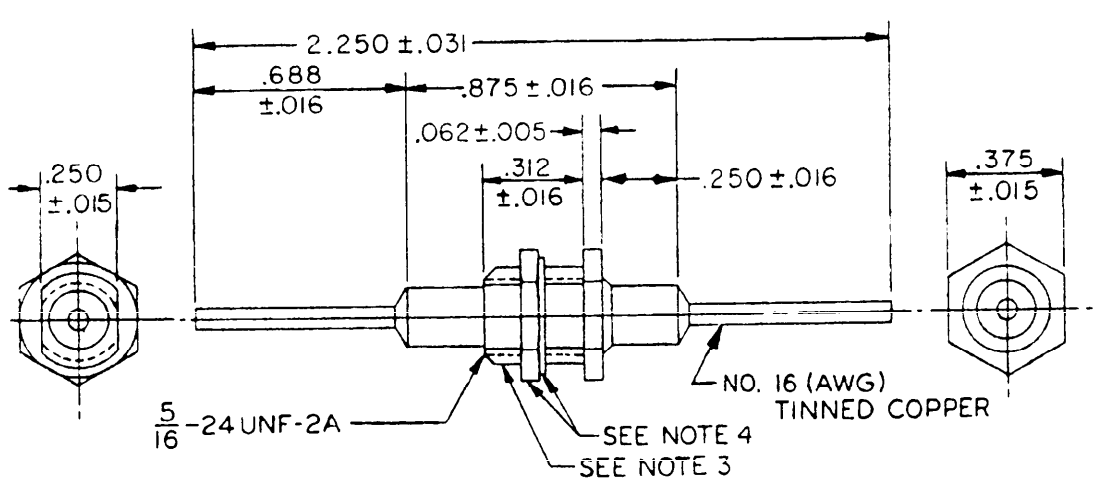
MIL-C-11015/23C
2 June 1982
SUPERSEDING
MIL-C-11015/23B
28 September 1970

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), (FEED-THRU), STYLE CK72

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this document and the latest issue of MIL-C-11015.



INCHES	MM
.005	.13
.015	.38
.016	.41
.031	.79
.062	1.57
.250	6.35
.312	7.92
.375	9.53
.688	17.48
.875	22.23
2.250	57.15



CIRCUIT DIAGRAM

NOTES:
1. All dimensions in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The threaded portion shall extend to within .047 (1.19 mm) of the shoulder.
4. A nut and internal-tooth lockwasher shall be supplied with each capacitor when requested by the procuring activity.

Figure 1. Dimensions and Configurations.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Feed-thru.
Capacitance value - See table I.
Capacitance tolerance - ±20% (M).
Rated temperature - -55° to +85°C.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 250% of rated voltage, applied between the case and the terminal.
Body insulation: Not applicable.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition B (50,000 ft).
Test potential - 1,550 volts dc.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 500,000 megohms, min, measured between either insulated point and capacitor case.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency:
100pF - 1 MHz ± 1 kHz.
Over 100 pF - 1 kHz ± 100 Hz.
Test voltage - 1.0 ± 0.2 volt rms.

Dissipation factor (DF): 1.2% max.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G's)

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TABLE I. Style CK72 characteristics.

Type designation	Rated voltage volts, dc	Rated temperature and voltage-temperature limits	Capacitance pF	Capacitance tolerance
CK72AX101M	1,500	AX	100	M
CK72AX471M	1,500	AX	470	M
CK72AX681M	1,500	AX	680	M
CK72AX102M	1,500	AX	1,000	M

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

Thermal shock and immersion:
Thermal shock: Method 107 of MIL-STD-202, condition A.
Immersion: Method 104 of MIL-STD-202, condition B.
IR - 200,000 megohms, min.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Terminal strength: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202.
DWV - 250% of rated voltage.
IR - 150,000 megohms, min.
Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202.
2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
IR - 500,000 megohms, min.
ΔCap. - ±5% of initial measurement.
ΔDF - 0.5%, max.

Voltage-temperature limits: AX; in accordance with MIL-C-11015.

Life (at elevated ambient temperature): Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).
Test potential - 200% of rated voltage.
IR - 200,000 megohms, min (at 85°C)
300,000 megohms, min (at 25°C).
DF - 1.5%, max (at 25°C).

Marking: The operating temperature range and voltage-temperature limit, capacitance, and capacitance tolerance shall be indicated on the capacitor by color coding as shown on figure 2 and table II. The MIL identifier (yellow spot) shall be applied to the capacitor body diametrically opposite the color spots. At the option of the manufacturer, capacitors may be marked in accordance with MIL-C-11015. In either case, full marking shall appear on the unit package.

TABLE II. Color code.

Color	Operating temperature range and voltage-temperature limits	Nominal capacitance		Capacitance tolerance
		First and second significant figures	Multiplier 1/	
Black.....	---	0	1	M
Brown.....	---	1	10	---
Red.....	AX	2	100	---
Orange.....	---	3	1,000	---
Yellow.....	---	4	---	---
Green.....	---	5	---	---
Blue.....	---	6	---	---
Purple (violet)	---	7	---	---
Gray.....	---	8	---	---
White.....	---	9	---	---

1/ The multiplier is the factor by which the two significant figures are multiplied to yield the nominal capacitance.

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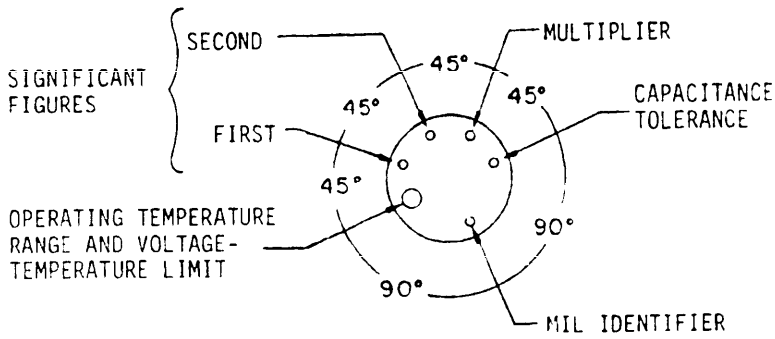


Figure 2. Color code marking.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, OS, SH
Air Force - 17, 99
DLA-ES

User activities:
Army -
Navy - MC
Air Force - 19

Agent:
DLA-ES

Preparing activity:
Navy - EC

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